



FOOD PACKAGING TECHNOLOGY

Subject Code: FDPT 313 | B.Tech Food Technology

Comprehensive Study on Materials, Permeability, and Advanced Systems

Core Functions of Packaging



Protection

Shields food from physical damage, chemical contamination, and biological spoilage during transit.



Containment

Encloses products to facilitate efficient transport, storage, and handling in various sizes.



Communication

Provides legal information, nutrition facts, branding, and usage instructions to consumers.

Paper and Paperboard

- ✓ **Versatility:** Lightweight and highly printable for branding.
- ✓ **Eco-Friendly:** Biodegradable and easily recyclable.
- ✓ **Types:** Includes Kraft paper, greaseproof paper, and corrugated fiberboard.
- ✓ **Limitation:** Poor moisture and gas barrier properties unless laminated or coated with plastics/wax.



Glass: The Inert Barrier



Premium Protection

Glass is highly inert, making it the safest contact material for acidic and high-moisture foods.

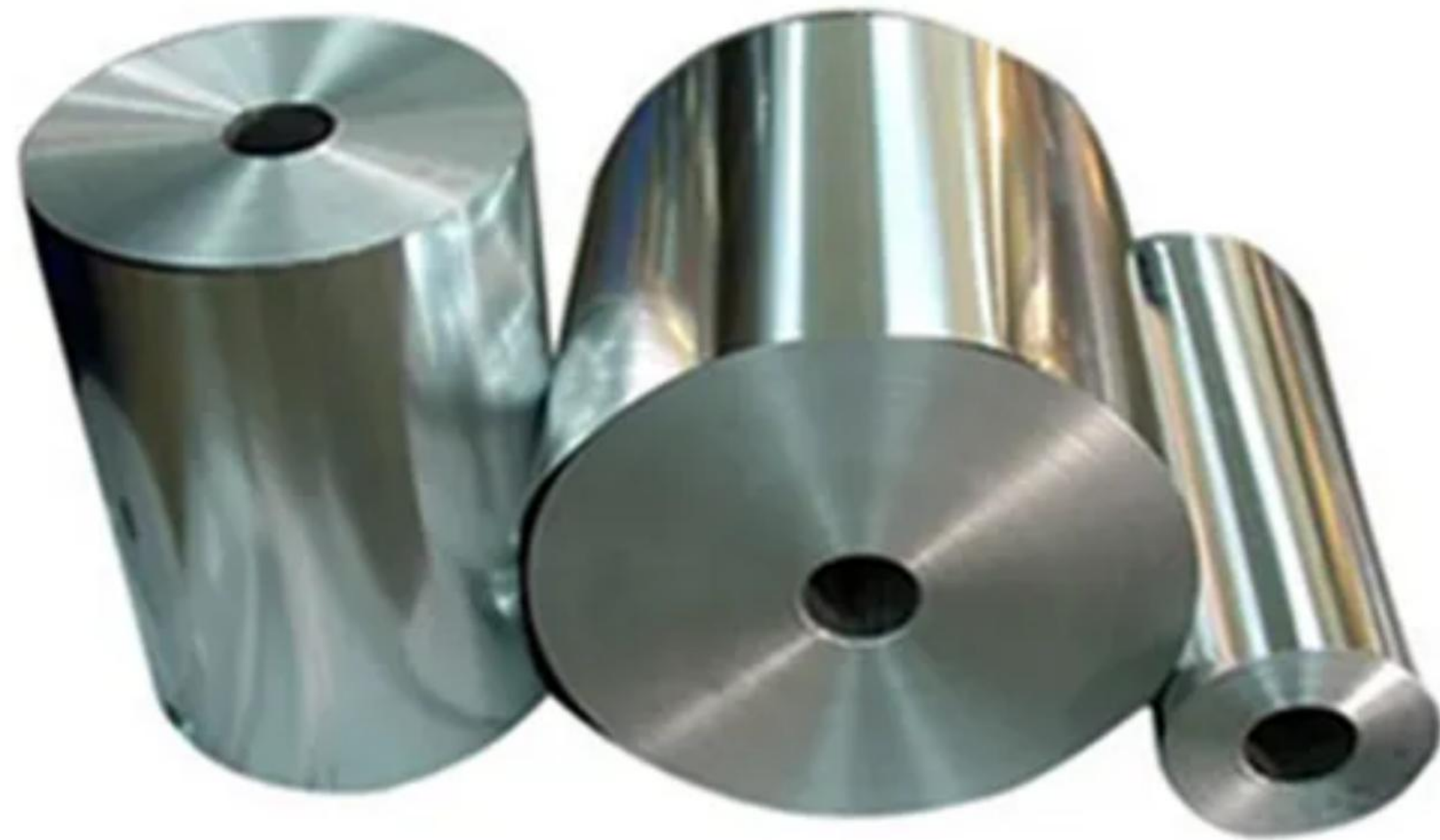
- ✓ **Impermeability:** Excellent barrier against gases and water vapor.
- ✓ **Transparency:** Allows consumers to inspect product quality visually.
- ✓ **Challenges:** Heavy weight increases shipping costs; fragility leads to breakage risks.

Metal Packaging Systems

Tinplate & Aluminum

Metallic containers offer total protection against light, air, and microorganisms, making them ideal for long-shelf-life canned goods.

- ✓ **Tinplate:** Steel coated with tin for corrosion resistance.
- ✓ **Aluminum:** Lightweight, flexible, and ideal for beverage cans and foils.
- ✓ **Mechanical Strength:** High resistance to heat processing (Canning/Retort).



Plastic Materials & Properties

 Plastic Containers

PET

Polyethylene Terephthalate:
High clarity, used for beverage bottles.

HDPE

High-Density Polyethylene:
Rigid and opaque, used for milk jugs.

LDPE

Low-Density Polyethylene:
Flexible, used for bread bags and wraps.

PP

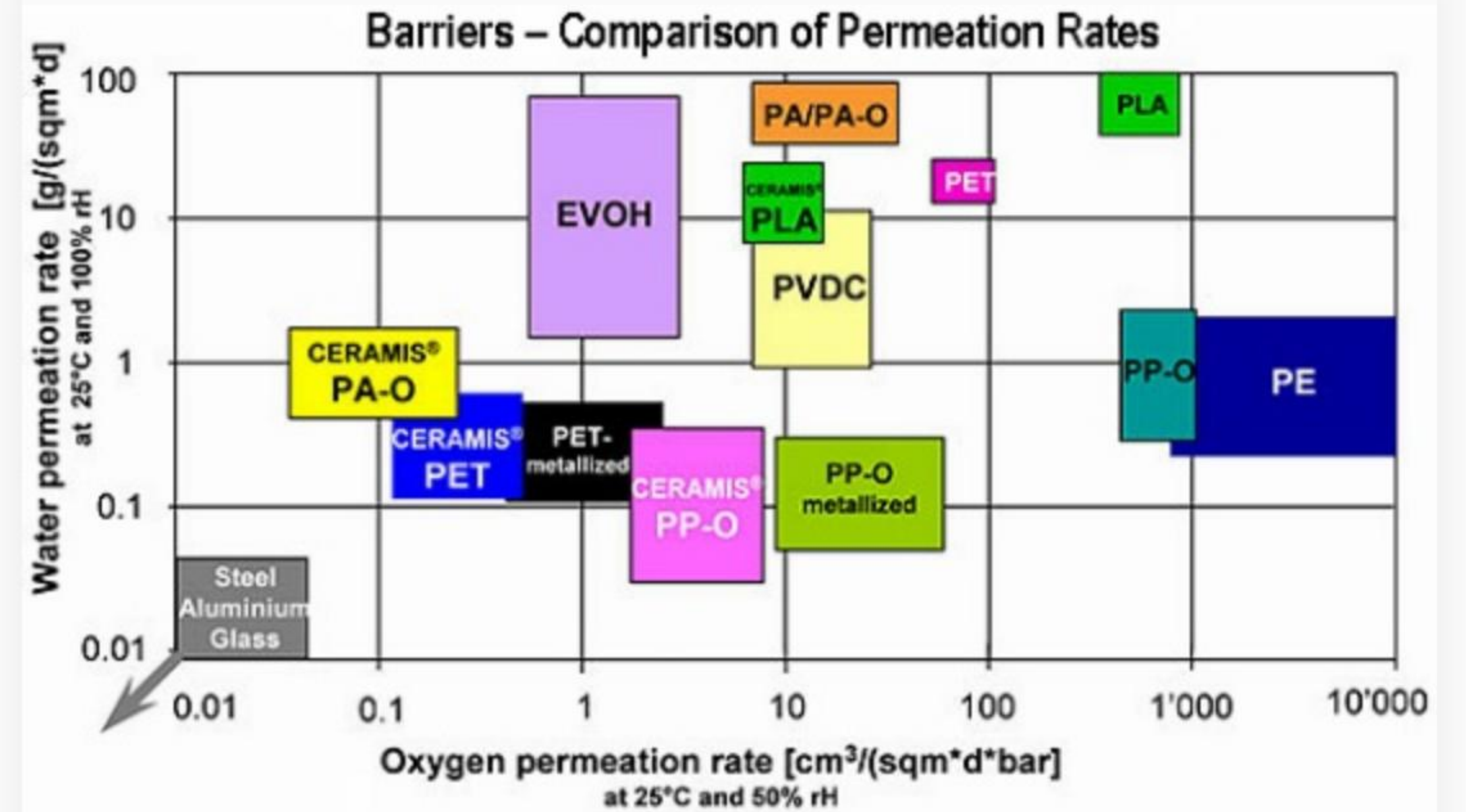
Polypropylene: High heat resistance, suitable for microwavable tubs.

Technical Concept: Permeability

Permeability determines how easily gases and vapors pass through a packaging material. Lower permeability equals better shelf-life.



Measured as OTR (Oxygen Transmission Rate) and WVTR (Water Vapor Transmission Rate).



Active Packaging Systems



Oxygen Scavengers

Removes residual oxygen in the headspace to prevent oxidation and rancidity in fats.



Moisture Absorbers

Controls relative humidity to prevent sogginess in dry snacks or mold growth in fresh produce.

Antimicrobials

Incorporates agents into the film to inhibit the growth of spoilage bacteria on surfaces.



technology adjusts the ambient atmosphere surrounding the food by removing some of the oxygen and replacing it with a mixture of carbon dioxide and nitrogen. This process – sometimes referred to as “gas flushing” – maintains the freshness and appearance of the food are preserved for longer periods of time, allowing for a more attractive product to be displayed on shelves and in stores.

Modified Atmosphere Packaging (MAP)

Replacing air inside a package with a protective gas mixture to extend fresh product life.

CO₂ (Carbon Dioxide)

Bacteriostatic effect; inhibits growth of aerobic bacteria and molds.

N₂ (Nitrogen)

Inert filler gas used to prevent package collapse and exclude oxygen.

The carbon dioxide and nitrogen mixture is more volatile to bacteria, yeasts, and other microorganisms that cause food spoilage and decay, meaning they can't wreak their usual havoc on food.



QUESTIONS?

Thank you for your attention

B.Tech Food Technology

Exam Preparation Study Resource - 2026

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